RILEY FRANCIS

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EDUCATION

University of Connecticut, Storrs, CT — Aug 2023 – Expected May 2027
Dual Degree: *BS Computer Science, BA Pure Mathematics*
Honors Program, Honors Scholar, COE Dean's List, CLAS Dean's List
GPA: 3.928 / 4.000

University of Edinburgh, Edinburgh, Scotland — Jan 2025 – May 2025
Visiting Student, School of Informatics

EXPERIENCE

SDE Co-op — Jun 2026 – Present
Amazon Robotics, Westborough, MA

- Member of Amazon Robotics Innovation Lab, developing perception, coordination, and interaction capabilities for a bi-manual package manipulation robot.
- Developed ROS2 packages and CI tools for our lab using Docker and Git.

Student Computer Programming Specialist — Apr 2023 – Jun 2025
University of Connecticut, Storrs, CT

- Handle redistribution and disposal of university electronic equipment.
- Configure and troubleshoot Windows, macOS, and Linux devices for reuse.

PROJECTS & RESEARCH

Reinforcement Learning on Autonomous Robots Inside Meta's Habitat-Lab — Aug 2025 – May 2026
Storrs, CT — Mentor: Dr. Jonathan Shihao Ji

- Developed a reinforcement learning method for training legged robots (i.e. Spot) inside Meta's Habitat-Lab.
- Collaborated in a team of four other researchers and formulated a research paper (available upon request).

Interpreting GNNs with MIPExplainer: Explaining Alzheimer's & Autism — May 2025 – Jul 2025
Greensboro, NC — Mentor: Dr. Chunjiang Zhu

- Applied MIPExplainer to brain GNNs to identify brain-region features influencing Alzheimer's and Autism prediction.
- Produced sparse model-level explanations to highlight key connectivity structures. Paper available upon request.

Making Lane Detection Efficient for Autonomous Model Cars — Feb 2021 – Jan 2023
Storrs, CT — Mentors: Dr. Jinbo Bi, Dr. Shengli Zhou, Dr. Caiwen Ding, Dr. Fei Miao

- Built and programmed autonomous model vehicles using AI and computer vision. Presented at IEEE/ACM ISLPED.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, HTML, CSS, JavaScript, TypeScript, React, LaTeX, SQL, PDDL
Packages: PyTorch, Torchvision, TensorFlow, Scikit-Learn, NetworkX, OpenCV, Gymnasium, NumPy, Matplotlib, Manim, SQLite, ROS2
Tools/Programs: Linux, Windows, Git, VS Code, Docker, AWS, Unity, Cloudflare, Bash, CMake

AWARDS & RECOGNITION

Honors Scholar (2023 – Present), CLAS Dean's List (2023 – Present), CoE Dean's List (2023 – Present), STEM Merit Scholarship (2023 – Present), 2025 Babbidge Scholar (2025), Global Citizen & Merit Scholarship (Sep 2024)

ACTIVITIES

Science Olympiad at UConn, Treasurer & Co-Founder — Aug 2023 – May 2026

- Planned and coordinated invitational competitions for dozens of high school and middle school teams.

RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Computer Vision, Robotics, Graph Theory, Algorithms